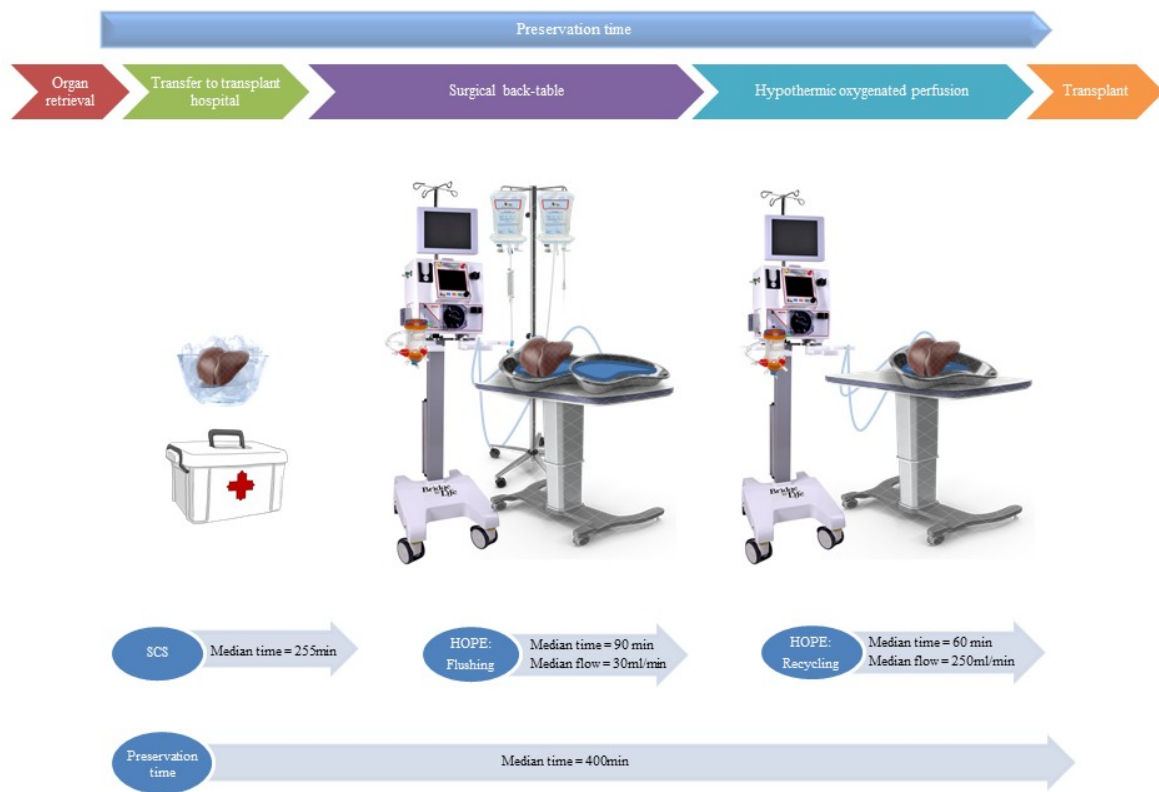


SUPPLEMENTAL MATERIAL

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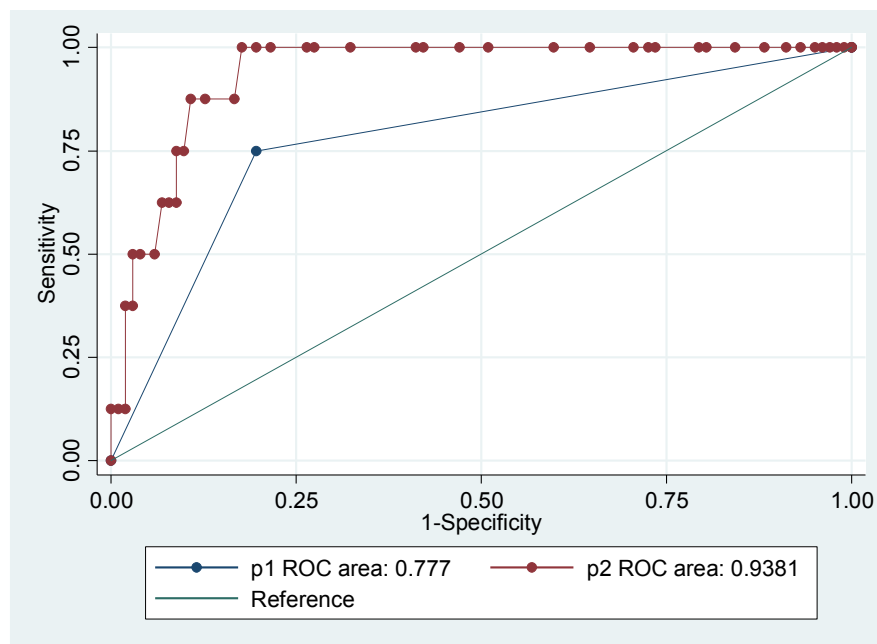
Protocol Algorithm for Liver Perfusion with HOPE



Supplemental Figure 1 The protocol algorithm for treatment of the study group undergoing preservation with hypothermic oxygenated perfusion (HOPE)

Comparison between EAD and EASE in predicting graft failure at 90 days

		Graft failure at 90 days		<i>RD (95% IC)</i>	<i>p value</i>	<i>AUROC</i>
		Yes	No			
EAD						
	yes – no. (%)	6 (23%)	20 (77%)	<i>-0.207 (-0.372 - -0.042)</i>	<i>0.003</i>	<i>0.777</i>
	no – no. (%)	2 (2%)	82 (98%)			
EASE						
	Median	-1.50	-3.50	<i>-0.532 (-0.901 - -0.164)</i>	<i>0.000</i>	<i>0.938</i>
	IQR	-2.15 - -1.00	-4.10 - -2.90			



Supplemental Figure 2 Comparison between EAD (blue line) and EASE (score predicting graft failure at 90 days

Predictive factors of EAD development among donor and recipient variables

Supplemental Table 2 Uni- and multivariate analysis of variables predicting EAD development				
Variable	Univariate logistic regression		Stepwise multivariate regression	
	RD (CI 95%)	P value	RD (CI 95%)	P value
HOPE	0.22 (0.07-0.37)	0.009	0.23 (0.09 – 0.38)	0.005
Age	-	<i>n.s.</i>		
Sex	-	<i>n.s.</i>		
BMI	-	<i>n.s.</i>		
Portal thrombosis	-	<i>n.s.</i>		
MELD score	-	<i>n.s.</i>		
Macrosteatosis	-0.01 (-0.15 - -0.01)	0.033	<i>n.s.</i>	
Microsteatosis	-0.06 (-0.01 - -0.01)	0.078	<i>n.s.</i>	
Fibrosi (Metavir)	-	<i>n.s.</i>		
Ishak Grade	-	<i>n.s.</i>		
DRI	0.33 (0.21 – 0.46)	0.079	<i>n.s.</i>	
CIT	-	<i>n.s.</i>		
Operation time	-0.01 (-0.003 - 0.01)	0.02	-0.01 (-0.01 - 0.001)	0.009

Predictive factors of EAD development among perfusate variables

Variable		Median (IQR)	Univariate logistic regression		Stepwise multivariate regression	
			RD (CI 95%)	P value	RD (CI 95%)	P value
Flushing						
pH	at T0 at T1	7.02 (6.98 – 7.07) 7.03 (6.97 – 7.07)	-	<i>n.s.</i>		
pCO2 (mmHg)	at T0 at T1	5 (5 – 5) 7.5 (6 – 9)	- -0.001 (-0.004 – 0.002)	<i>n.s.</i> 0.009	<i>n.s.</i>	
pO2 (mmHg)	at T0 at T1	245 (229 – 254) 202 (170 – 259)	-	<i>n.s.</i>		
Glucose (mg/dL)	at T0 at T1	6 (3 – 12) 147 (121 – 176)	- -0.002 (-0.005 – 0.001)	<i>n.s.</i> 0.053	<i>n.s.</i>	
Lactate (mmol/L)	at T0 at T1	0.2 (0.2 - 0.3) 2.3 (1.9 – 2.8)	- -0.011 (-0.285 – 0.007)	<i>n.s.</i> 0.012	-0.011 (-0.285 - 0.007)	0.012
Recycle						
pH	at T0 at T1	7.17 (7.11 – 7.24) 7.12 (7.09 – 7.16)	-	<i>n.s.</i>		
pCO2 (mmHg)	at T0 at T1	5 (5 – 6.5) 6 (5 – 7)	-	<i>n.s.</i>		
pO2 (mmHg)	at T0 at T1	271 (183 – 454) 354 (234 – 520)	-	<i>n.s.</i>		
Glucose (mg/dL)	at T0 at T1	194 (187 – 210) 254 (209 – 293)	-	<i>n.s.</i>		
Lactate (mmol/L)	at T0 at T1	1.55 (0.65 – 2.25) 3.05 (2.25 – 4.20)	- 0.011 (-0.025 – 0.003)	<i>n.s.</i> 0.014	<i>n.s.</i>	